AutoEncoder-Compressed Parallel Split Learning for Pre-trained Model Fine-Tuning

Bas Meuwissen · **Vasileios Tsouvalas** · Nirvana Meratnia

Eindhoven University of Technology

4th Workshop on Advancements in Federated Learning (WAFL), ECML-PKDD 2026

01 Motivation

The problem

Split learning moves expensive computation to a server, but every training step still transfers activations and gradients. Compression must preserve the feature distribution learned during pretraining.

Contributions

- **Bidirectional compression:** reduce activation and gradient traffic through an aligned autoencoder (AE).
- **Client adaptation:** retain local encoders while sharing one averaged server decoder.
- **Deployment evaluation:** test global and local inference across four vision benchmarks.

Pretrained features are fragile

A randomly initialized AE can destroy useful representations. The compressor must match the pretrained feature distribution before downstream fine-tuning.

Two deployment settings

Global inference removes the AE; local inference retains it. The first three benchmark datasets use global evaluation, while FEMNIST uses local evaluation.

Scope

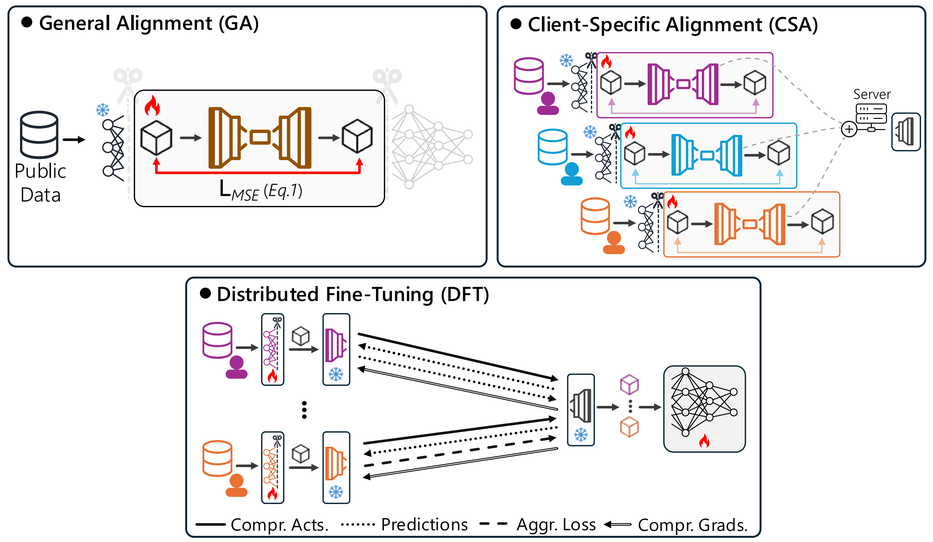
Evaluated on supervised vision tasks with vision transformer (ViT)-B/32; multimodal and language adaptation remain untested. Public-data alignment and client warm-up are required.

Notation

parallel split learning (PSL) · mean squared error (MSE) · standard deviation (SD)

02 AE-PSL

Align the compressor before fine-tuning



Technical core

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} \left\| H(x) - D_\psi(E_\phi(H(x))) \right\|_2^2$$

H: frozen pretrained features; E/D: encoder/decoder. Minimize reconstruction error on public data, then each client's data.

Paper

Code

03 Evaluation

≈10×
communication reduction
Relative to uncompressed split fine-tuning.

82.7 ± 0.2%
CIFAR-100 accuracy
5 clients; matches the uncompressed reference.

Accuracy at approximately 10× communication reduction

Method	CIFAR-100	Food101	SUN397	FEMNIST
No compression (R=1)	82.7 ± 0.2	69.1 ± 0.9	67.0 ± 0.6	87.9 ± 0.3
C3-SL	50.9 ± 1.2	40.3 ± 1.2	41.3 ± 2.1	83.5 ± 0.1
Rand-Top-K	76.2 ± 0.4	60.2 ± 0.2	62.9 ± 1.0	87.2 ± 0.4
ADC	68.4 ± 0.4	54.6 ± 1.6	52.9 ± 1.1	81.1 ± 0.9
AE-PSL (Ours)	**82.7 ± 0.2**	**69.6 ± 1.2**	**66.7 ± 0.3**	**88.0 ± 0.1**

Top-1 accuracy (%), mean ± SD, 3 seeds; ViT-B/32, 5 clients. Compressed methods R≈10. Global: first three datasets, AE bypassed; local: FEMNIST, AE active.

Alignment prevents feature mismatch

Ablation	CIFAR-100	FEMNIST
Unaligned AE	1.4 ± 0.3	86.9 ± 2.0
+ General alignment	75.2 ± 0.4	87.1 ± 1.1
+ Freeze AE	79.6 ± 0.9	88.0 ± 0.6
+ Client alignment (Ours)	**82.2 ± 0.7**	**88.0 ± 0.7**

Cumulative ablation; accuracy (%), reported ±, 3 seeds; ViT-B/32, 5 clients, R=15. CIFAR-100: global; FEMNIST: local.

Takeaway

AE-PSL matches uncompressed CIFAR-100 accuracy at **≈10× lower communication**. **Alignment, freezing and client adaptation** each improve the reported CIFAR-100 ablation.

On efficacy of Meta-Learning for Domain Generalization in Speech Emotion Recognition

Raeshak King Gandhi · **Vasileios Tsouvalas** · Nirvana Meratnia

Eindhoven University of Technology

PerFail · IEEE PerCom Workshops 2023 · pp. 421-426

01 Motivation

The problem

Speech emotion recognition (SER) accuracy often drops across speakers, languages and corpora. Can training across source domains improve performance on a target corpus that is entirely unavailable during training?

Contributions

- **Domain-shift weighting** adjusts the meta-test loss using an energy-score gap.
- **Seven transfer scenarios** compare conventional training, meta-learning and feature representations.
- **Negative-result analysis** identifies persistent barriers to practical cross-corpus emotion recognition.

Target stays unseen

Training uses source corpora only. The target corpus is reserved for evaluation, distinguishing domain generalization (DG) from adaptation using target-domain samples.

Why audio is difficult

Emotion corpora are small and differ in speakers, language and class coverage. These differences make cross-corpus comparisons difficult and can obscure shared emotional features.

Scope

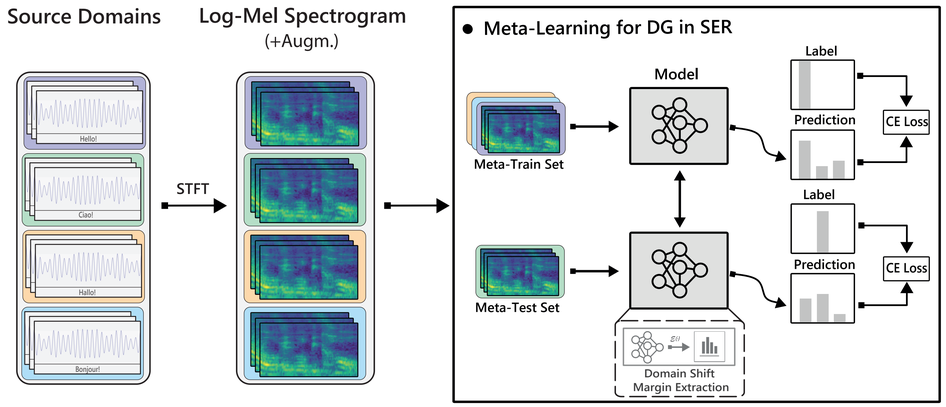
Gains remain uneven and insufficient for practical use. Small corpora, missing classes and unreported trial dispersion limit the strength of the comparisons.

Notation

domain-shift aware meta-learning for domain generalization (*DA-MLDG*) · short-time Fourier transform (STFT) · cross-entropy (CE)

02 DA-MLDG

Weight source-domain shifts to test unseen-corpus transfer



Technical core

$$\mathcal{L} = \mathcal{L}_{\theta}(\mathcal{S}') + \gamma \mathcal{L}_{\theta'}(\mathcal{V})$$

S′ and V are disjoint source subsets. θ′ follows a meta-train update; γ controls the weight of its meta-test loss.

$$\gamma = |\mathcal{E}(\mathcal{S}', p_{\theta'}) - \mathcal{E}(\mathcal{V}, p_{\theta'})|$$

E is the absolute mean log-sum-exp of temperature-scaled logits. A larger source-domain gap gives the meta-test loss more weight.

Paper

Code

03 Evaluation

4 / 7
above supervised Deep-All
Log-mel transfer scenarios

2 / 7
highest among four methods
Log-mel scenarios 4 and 5

Cross-domain emotion recognition

Scenario / target	Deep-All	RSC	MLDG	DA-MLDG (Ours)
1 / IEMOCAP	26.95	32.36	27.61	**31.35**
2 / RAVDESS	43.40	44.78	38.86	**39.41**
3 / TESS	47.03	43.77	46.29	**46.80**
5 / AESDD	60.65	56.69	60.45	**62.26**
6 / ShEMO	31.54	40.70	37.87	**38.79**

Held-out target accuracy (%), ten-trial mean; log-mel inputs with a VoxCeleb-pretrained ResNet-18.

Transfer with pretrained audio embeddings

Scenario / target	Deep-All (%)	DA-MLDG (%) (Ours)
1 / IEMOCAP	28.57	**32.11**
2 / RAVDESS	42.61	**39.82**
5 / AESDD	61.17	**64.11**
6 / ShEMO	33.16	**40.22**

Same source/target scenarios with TRILLsson embeddings; ten-trial mean target accuracy (%).

Takeaway

Transfer improves unevenly: *DA-MLDG* beats supervised Deep-All in four of seven log-mel scenarios and leads all four methods in two. With pretrained embeddings, it beats the supervised baseline in **only three of seven** scenarios.

DeltaMask: Minimizing Communication in Federated Fine-Tuning of Vision Foundation Models

Vasileios Tsouvalas · Yuki M. Asano · Aaqib Saeed

Eindhoven University of Technology, The Netherlands | University of Technology Nuremberg, Germany

IEEE BigData 2024

01 Motivation

The problem

Fine-tuning large vision models across clients creates substantial communication costs. Even one-bit updates can be expensive, so adaptation must preserve useful predictions while sending less information each round.

Contributions

- **Frozen-backbone adaptation:** Learn task-specific stochastic masks for pretrained vision models.
- **Selective updates:** Rank mask changes and compress their indices with probabilistic filters.
- **Empirical comparison:** Evaluate accuracy and bitrate across eight datasets and multiple pretrained architectures.

Federated adaptation setting

In federated learning (FL), clients adapt a shared pretrained model without collecting their images centrally. Model size drives communication cost, even when pretrained representations already transfer well to local classification tasks.

Difference from complete-mask coding

FedPM compresses complete sampled masks with arithmetic coding. *DeltaMask* instead exploits changes relative to a reproducible server mask; communication savings must be assessed together with classification accuracy.

Scope

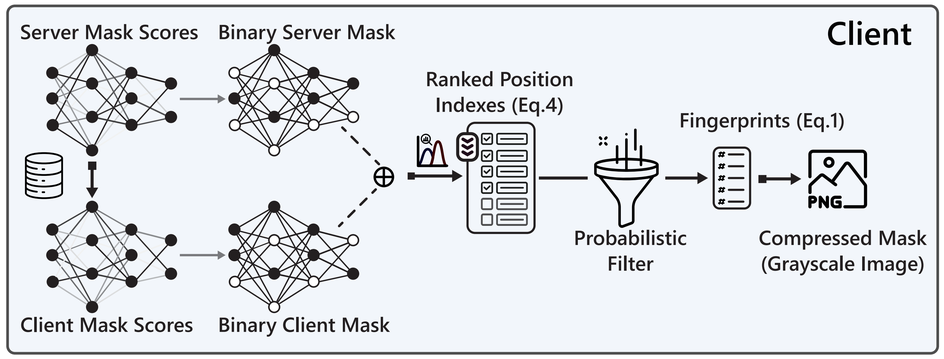
Probabilistic filters can introduce false-positive mask changes. Experiments cover image classification, and accuracy remains below unrestricted fine-tuning.

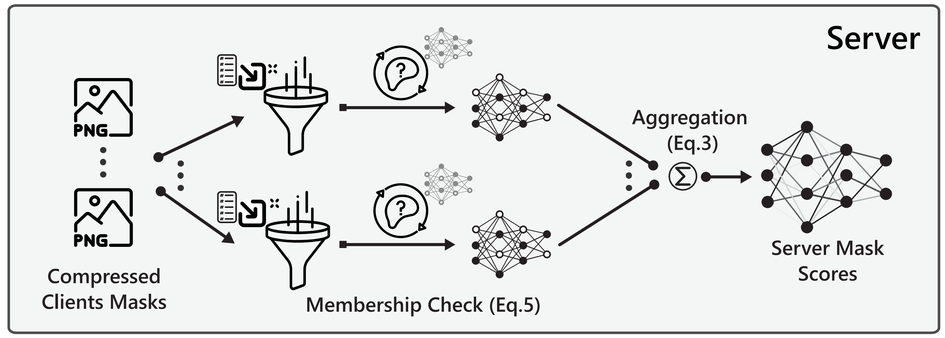
Notation

Dirichlet distribution with concentration a (Dir(a)) · independent and identically distributed (IID) · Kullback-Leibler divergence (KL) · bits per parameter (bpp) · vision transformer (ViT) · percentage points (pp)

02 DeltaMask

Communicate selected mask changes





Technical core

$$\widetilde{w} = m \odot w_{\text{pre}}, \qquad m_i \sim \text{Bernoulli}(\theta_i)$$

w_{pre}: frozen pretrained weights; θ: learned keep probabilities; m: sampled binary mask. Adapt which weights are active.

$$\Delta = \{i : m_i^{\text{local}} \neq m_i^{\text{server}}\}, \qquad \Delta' = \text{TopK}_{\text{KL}}(\Delta)$$

Δ: changed mask indices; K: retained count. Rank changes by local-to-server Bernoulli KL divergence, then send selected indices.

Paper

Code

03 Evaluation

0.151 bpp
Average transmitted bitrate
8 datasets · CLIP ViT-B/32 · IID

85.44%
Average accuracy
Same setting · full participation

Low bitrate with a measured accuracy trade-off

Method	Avg. accuracy (%)	Avg. bits/parameter
Fine-tuning	88.08	32
FedMask	79.23	1.0
EDEN	84.14	0.691
FedPM	86.07	0.872
DeltaMask (Ours)	**85.44**	**0.151**

CLIP ViT-B/32 · 8 datasets · IID Dir(10) · 30 clients · full participation · 100 rounds. Means of 3 runs.

Non-IID data with 20% participation

Method	Avg. accuracy (%)	Avg. bits/parameter
Fine-tuning	83.19	32
EDEN	78.56	0.715
FedPM	81.64	0.948
DeltaMask (Ours)	**81.13**	**0.233**

Same 8 datasets and backbone · non-IID Dir(0.1) · 30 clients · 20% participation · 300 rounds. Means of 3 runs.

Takeaway

DeltaMask sends **selected mask changes**, reaching **0.151 bits/parameter** at **85.44% accuracy** across eight IID tasks. Compared with FedPM at 0.872 bits/parameter, it reduces communication while trailing accuracy by 0.63 pp.

EFU: Enforcing Federated Unlearning via Functional Encryption

Samaneh Mohammadi · **Vasileios Tsouvalas** · Iraklis Symeonidis · Ali Balador · Tanir Ozcelebi · Francesco Flammini · Nirvana Meratnia

RISE Research Institutes of Sweden | Mälardalen University | Eindhoven University of Technology

ACM CIKM 2025

01 Motivation

The problem

A client's unlearning update is ineffective if omitted from aggregation. *EFU* addresses this protocol gap while studying how forgetting affects retained utility and observable model changes.

Contributions

- **Aggregate authorization:** Bind encrypted updates to an agreed aggregation function.
- **Local unlearning:** Combine forget loss, adversarial training and inverted Memory Aware Synapses (MAS) regularization.
- **Forgetting evaluation:** Compare sample-, class- and task-level removal with full retraining.

Threat model

The server is honest-but-curious (HbC) and may inspect updates or try to omit an unlearning contribution. Clients are trusted to perform key generation, encryption and unlearning correctly; no trusted third-party authority (TPA) is assumed.

Protection boundary

Functional keys authorize one round's aggregate. Update-type hiding assumes indistinguishable outputs and excludes full collusion; visible mappings and side channels remain separate concerns. This does not compel deployment or certify complete data erasure.

Scope

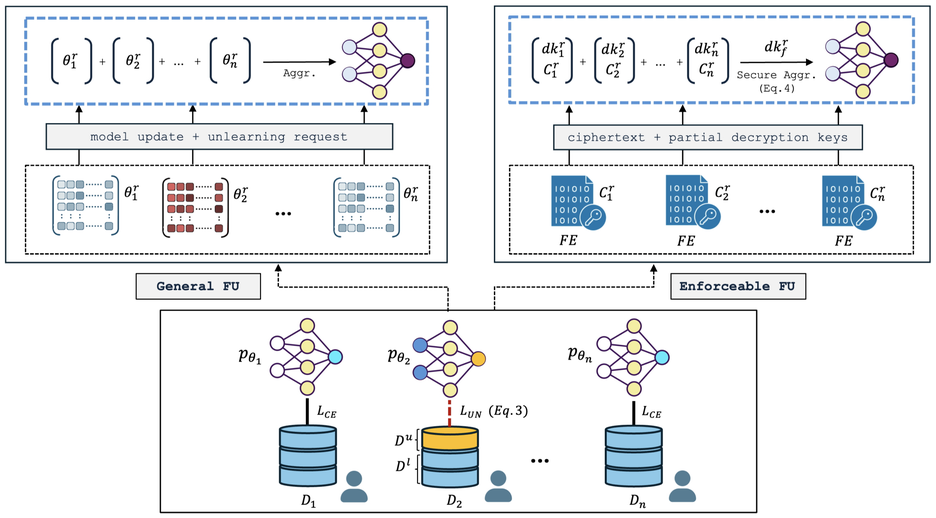
Results average three runs without reporting dispersion. Similar retained/forgotten accuracies do not establish statistical equivalence to retraining.

Notation

Dirichlet distribution with concentration a (Dir(a)) · Labeled Faces in the Wild (LFW) · functional encryption (FE) · federated unlearning (FU) · vision transformer (ViT) · unlearning loss (UN) · cross-entropy (CE)

02 EFU

Bind updates to the authorized aggregate



Technical core

$$\mathcal{L}_{\text{UN}} = -\mathcal{L}_{\text{CE}}(D^u) + \mathcal{L}_{\text{CE}}(D^{u,\text{adv}}) + \sum_j \bar{\Omega}_j(\theta_j - \widetilde{\theta}_j)^2$$

D^u: forget data; $D^{u,adv}$: adversarial variants; $\tilde{\theta}$: starting weights; $\bar{\Omega}$: inverted MAS importance. Task forgetting also optimizes retained-task loss.

$$\theta_j^{r+1} = \text{Dec}(\{\theta_{i,\text{enc}}^r[j]\}_i, \text{dk}_f)$$

Encrypted client updates and the function key dk_f yield the agreed aggregate. The binding concerns decryption, not subsequent deployment.

Paper

Code

03 Evaluation

95.37%
Retained gender accuracy
LFW · task forgetting

5.12%
Forgotten identity accuracy
Same *EFU* model

Class forgetting with retained utility

Method	Retain accuracy (%)	Forget accuracy (%)
Full retrain	94.82	13.72
PGD	91.82	14.31
FedOSD	93.06	12.77
EFU + FedOSD (Ours)	**92.73**	**12.54**
EFU (Ours)	**92.43**	**13.06**

CIFAR-10 · ResNet-18 · forget 10% of classes · 10 clients (1 unlearning) · 20% participation · Dir(0.1). 100 rounds; 10 unlearning rounds from round 50. Means of 3 runs; higher retain / lower forget is better.

Forget identity, retain gender

Method	Gender accuracy (%)	Identity accuracy (%)
Full retrain	96.86	4.71
PGD	94.81	10.32
FedOSD	95.02	5.01
EFU (Ours)	**95.37**	**5.12**

LFW · ViT-B/32 · 30 clients, all unlearning identity · full participation. Ten unlearning rounds from round 50; means of 3 runs.

Takeaway

EFU retains **95.37% gender accuracy** while identity accuracy falls to **5.12%** in LFW task forgetting. Its **authorized encrypted aggregation** incorporates the agreed updates, while complete erasure and subsequent model deployment remain separate requirements.

EncCluster: Scalable Functional Encryption in Federated Learning through Weight Clustering and Probabilistic Filters

Vasileios Tsouvalas · Samaneh Mohammadi · Ali Balador · Tanir Ozcelebi · Francesco Flammini · Nirvana Meratnia

Eindhoven University of Technology, The Netherlands | RISE Research Institutes of Sweden, Sweden | Mälardalen University, Sweden

Pervasive and Mobile Computing, 2025

01 Motivation

The problem

Secure federated learning (FL) must hide individual model updates without making communication and client computation impractical. Encrypting every weight is costly, especially as neural networks grow.

Contributions

- **Compact encryption:** Combine weight clustering with decentralized multi-client functional encryption (DMCFE).
- **Mapping compression:** Encode weight-to-centroid assignments with Binary Fuse (BF) filters.
- **Measured trade-offs:** Evaluate accuracy, uploads and encryption time across four architectures.

Threat model

An honest-but-curious (HbC) server aggregates updates, while an HbC third-party authority (TPA) initializes the cryptosystem; both may seek private client information. Clients may collude by sharing their own weights. Secure channels, protected key provisioning and stable connections are assumed.

Protection boundary

DMCFE provides aggregate-only decryption under the stated key assumptions. The server recovers position mappings, so these are not secret from it; BF filters do not inherit cryptographic guarantees. Poisoning and denial-of-service are outside the analysis.

Scope

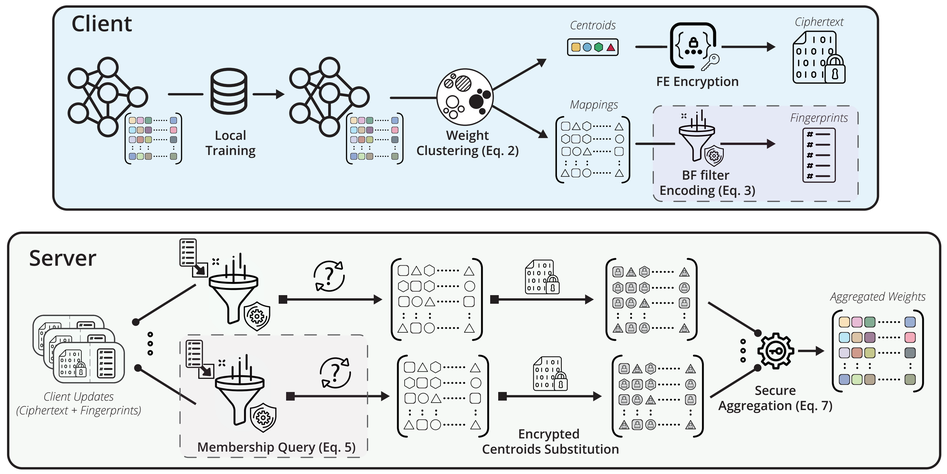
Timings come from a cluster simulation and are hardware-specific; repeated-run uncertainty is unreported. Filter errors do not establish differential privacy.

Notation

Dirichlet distribution with concentration a (Dir(a)) · independent and identically distributed (IID) · functional encryption (FE) · percentage points (pp)

02 EncCluster

Encrypt centroids instead of every model weight



Technical core

$$\mathcal{L}_{\text{cluster}} = \sum_{i=1}^{d} (\theta_i - z_{a_i})^2$$

θ_i: weight; z: centroids; a_i: assigned cluster. Approximate many weights with a few shared values before encryption.

$$\theta_j^{r+1} = \frac{1}{|D|} \text{Dec}(\{\widehat{\theta}_{n,j}^{r}\}_n, \text{dk}_f)$$

Encrypted weights are preweighted by client sample count. Divide the decrypted sum by total samples |D|; dk_f authorizes aggregation.

Paper

Code

03 Evaluation

0.284×
Upload volume / FedAvg
CIFAR-10 · full participation

2.04 s
Client encryption time
A10 / 96-core cluster simulation

Lower encrypted communication and computation

Method	IID Δ (pp)	Non-IID Δ (pp)	Upload / FedAvg	Encrypt (s)
FedAvg + clustering	-0.12	-0.71	0.034	—
DeTrust-FL	-0.09	-0.10	3.743	329.53
SEFL	-0.98	-1.74	2.581	7.64
EncCluster (Ours)	**-0.32**	**-0.79**	**0.284**	**2.04**

CIFAR-10 · ResNet-20 · 30 clients · full participation · 100 rounds. Dir(10)/Dir(0.1), 128 centroids, 256-bit keys. Deltas vs. FedAvg; timings from an A10 / 96-core cluster simulation.

Architecture coverage

Architecture	C10 Δ (pp)	C100 Δ (pp)	Encrypt (s)
ConvNeXt-Tiny	-0.36	-1.29	2.02
ResNet-20	-0.32	-1.19	2.03
MobileNet	-0.24	-0.97	2.08
ConvMixer-256/8	-0.19	-1.02	2.12

IID CIFAR-10/100 · 10 clients · full participation · 100 rounds. Accuracy changes use each architecture's FedAvg baseline; same cluster hardware.

Takeaway

EncCluster reduces uploads to **0.284× FedAvg** with a **0.32-point IID accuracy decrease** on CIFAR-10. Client encryption takes **2.04 s** in the A10 / 96-core cluster simulation, versus 7.64 s for SEFL.

FedCode: Communication-Efficient Federated Learning via Transferring Codebooks

Saeed Khalilian · **Vasileios Tsouvalas** · Tanir Ozcelebi · Nirvana Meratnia

Eindhoven University of Technology, The Netherlands

IEEE EDGE 2024

01 Motivation

The problem

Clustering compresses weight values, but full index maps still scale with model size. Repeatedly transferring those assignments can dominate communication, while stale mappings can prevent convergence.

Contributions

- **Bidirectional savings:** exchange codebooks for both server-to-client and client-to-server communication.
- **Index-free rounds:** update weights by nearest-center mapping without transmitting assignments.
- **Periodic calibration:** restore assignments to manage the communication-accuracy trade-off.

Two parts of a compressed model

A codebook stores K cluster centers; an index map assigns P weights to them. Sending centers alone avoids the P-sized assignment payload.

What the reduction measures

Data transmission reduction (DTR) is FedAvg bits divided by method bits. Larger values mean less communication; they do not establish latency, energy or client wall-clock savings.

Scope

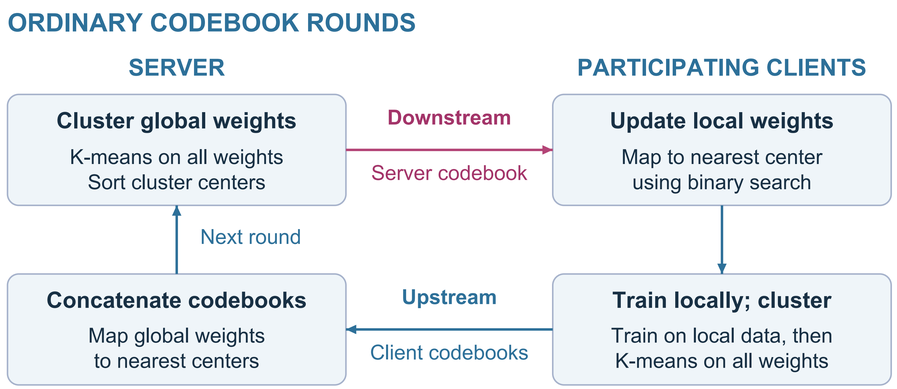
F1 and F2 control periodic index exchange; clustering adds local work. Calibration is needed for the reported accuracy, and repeated-run uncertainty and end-to-end runtime are unreported.

Notation

independent and identically distributed (IID)

02 FedCode

Codebook exchange with periodic calibration



Codebook = cluster centers; ordinary rounds omit weight indices.

PERIODIC CALIBRATION — SELECTED ROUNDS

Transfer codebook + weight indices in the relevant direction.

Clients reconstruct weights from indices and centers.

The server reconstructs client models, then applies FedAvg.

Technical core

$$j^* = \arg\min_j |w_i - c_j|, \qquad w_i \leftarrow c_{j^*}$$

w_i: existing weight; c_j: received center. Ordinary rounds choose the nearest center; the server uses concatenated client codebooks.

$$\mathrm{DTR} = \frac{2Pb}{2Kb + (F_1 + F_2)P\lceil \log_2 K \rceil}$$

P weights, K centers, b bits/weight; F1 and F2 are calibration frequencies. Less index transfer reduces traffic.

Paper Code

03 Evaluation

14.6×
joint communication reduction
IID CIFAR-100; ResNet-20.

58.07%
FedCode test accuracy
FedAvg: 61.22% on the same workload.

Compress both communication directions

Method	Accuracy (%)	Down DTR	Up DTR	Joint DTR
FedAvg	61.22	—	—	—
OMC-T	61.21	—	2.9	1.4
FedCompress	59.67	4.11	4.11	4.11
FedZip	57.98	—	19.8	1.9
FedCode (Ours)	**58.07**	**24.2**	**10.4**	**14.6**

IID CIFAR-100 · ResNet-20 · 10 clients · full participation · 100 rounds · 4 local epochs. 64 centers; F1=0.2, F2=0.5.

Calibration controls the trade-off

F1 / F2	Accuracy Δ (pp)	Joint DTR	Data (MB)
1 / 1	-0.86	5.3	394.21
0.2 / 0.5	-3.15	14.6	143.92
0.1 / 0.1	-21.64	44.3	47.35
0 / 0	-46.88	264.7	7.9

Same setting; accuracy change vs FedAvg. FedAvg communication: 2102 MB.

Takeaway

FedCode achieves **14.6× joint communication reduction** at 58.07% accuracy, 3.15 points below FedAvg. Without calibration, communication falls further but accuracy drops **46.88 points** below FedAvg.

Communication-Efficient Federated Learning through Adaptive Weight Clustering and Server-Side Distillation

Vasileios Tsouvalas · Aaqib Saeed · Tanir Ozcelebi · Nirvana Meratnia

Eindhoven University of Technology

IEEE ICASSP 2024

01 Motivation

The problem

Repeated model exchange makes federated learning (FL) expensive on constrained devices. Compressing client weights alone offers little download savings because aggregation breaks their shared-centroid structure.

Contributions

- **Bidirectional compression** reduces uploads and downloads without changing the aggregation rule.
- **Adaptive cluster capacity** tracks local representation diversity through effective rank.
- **Five-task evaluation** measures accuracy, traffic and model size across vision and audio.

Aggregation mismatch

Different clients learn different centroids. Averaging their weights disrupts this structure, so client compression alone does not preserve a compact broadcast model.

Available data

Clients hold labeled samples and a small unlabeled validation set. The server requires auxiliary out-of-distribution (OOD) data, which need not match client distributions.

Scope

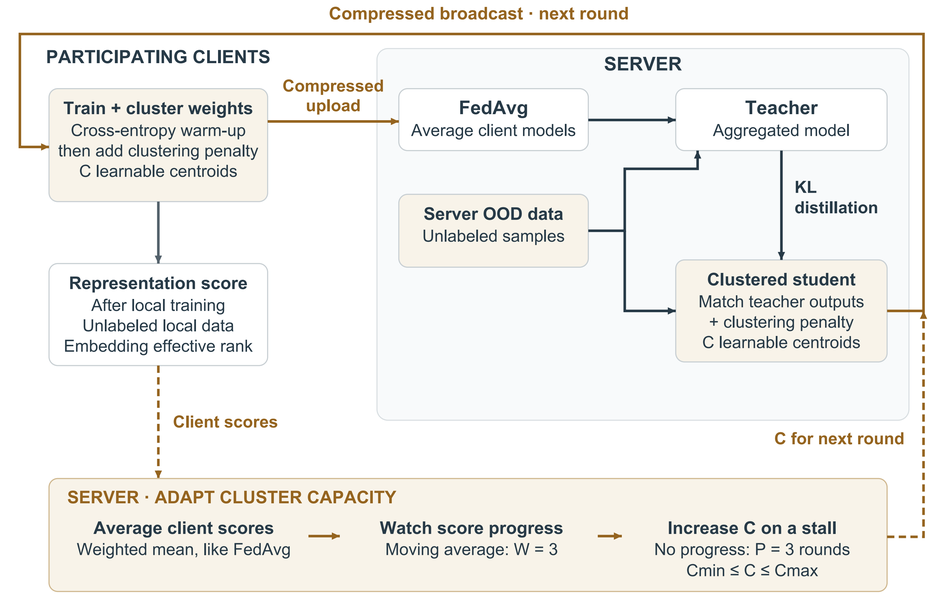
Server distillation adds optimization work. Evaluation uses simulated clients and two-run accuracy means without reported dispersion.

Notation

Kullback-Leibler divergence (KL) · weight clustering (WC)

02 FedCompress

Compress uploads and downloads with adaptive cluster capacity



Technical core

$$\mathcal{L}_S = \mathcal{L}_{\mathrm{KL}}(p_T \| p_\theta) + \beta \mathcal{L}_{\mathrm{wc}}(\theta, \mu, C)$$

Teacher p_T guides student p_θ; the clustering penalty groups weights around C centroids μ. β balances compression and imitation.

$$\mathcal{E}_k = \exp\left(-\sum_j r_j \log r_j\right)$$

r_j are normalized singular values of client k's embeddings. Their effective rank tracks representation diversity and guides cluster capacity.

Paper

Code

03 Evaluation

5.41×
communication reduction
VoxForge versus FedAvg

-0.31 pp
accuracy difference
Same VoxForge comparison

Communication savings with small accuracy losses

Dataset	FedAvg (%)	Δ acc. (pp)	Comm. (×)	Model (×)
CIFAR-10	86.26	-1.83	4.53	5.18
CIFAR-100	60.68	-1.88	3.80	3.93
PathMNIST	88.22	-1.72	4.79	5.27
SpeechCommands	95.75	-0.42	5.04	5.09
VoxForge	81.05	-0.31	5.41	5.64

20 clients, 20 rounds, 10 local/server epochs; two-run mean accuracy. Signed accuracy changes and reduction ratios are relative to FedAvg.

Why compress again at the server?

CIFAR-100 method	Δ acc. (pp)	Comm. (×)	Model (×)
FedZip	-2.57	1.94	2.11
Without server compression	-2.67	1.02	1.62
FedCompress (Ours)	**-1.88**	**3.80**	**3.93**

CIFAR-100, same protocol; accuracy changes and reduction ratios relative to FedAvg.

Takeaway

Communication falls across all five tasks, with small accuracy losses. On CIFAR-100, **server compression** raises the communication reduction from 1.02× to 3.80×.

Labeling Chaos to Learning Harmony: Federated Learning with Noisy Labels

Vasileios Tsouvalas · Aaqib Saeed · Tanir Ozcelebi · Nirvana Meratnia

Eindhoven University of Technology, The Netherlands

ACM TIST · Vol. 15, No. 2, Article 22 · 2024

01 Motivation

The problem

Federated learning (FL) must learn from unreliable client labels despite uneven noise, limited supervision and unequal device resources.

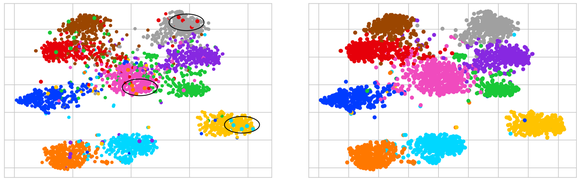
Contributions

- **Client-specific noise estimates** identify uneven supervision quality across devices.
- **Three alternative interventions** match correction, teacher guidance or aggregation to device resources.
- **Broad evaluation** tests synthetic vision/audio noise and human-annotated image noise.

Correct labels before training

Nearest neighbor-based correction (*NNC*) uses local embedding neighbors to correct class labels before training. Input examples and embeddings stay fixed.

Before and after label correction



CIFAR-10; 40% random noise. Fixed contrastive language-image pre-training (CLIP) features.

Scope

Pretrained encoders add cost; compute is unmatched and three-run results omit dispersion. *NNC* can fall below FedAvg under class flipping.

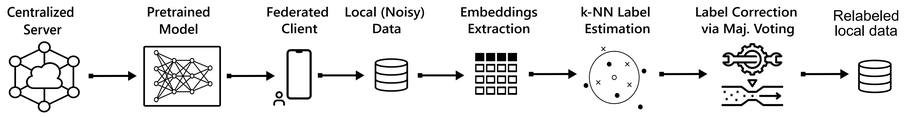
Notation

adaptive knowledge distillation (*AKD*) · noise-aware federated averaging (*NA-FedAvg*) · knowledge distillation (KD) · k-nearest neighbors (kNN) · confident learning (CL) · percentage points (pp) · cross-entropy (CE)

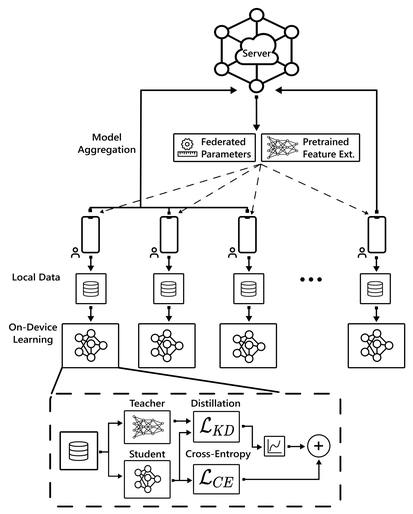
02 FedLN

Three alternative interventions address client label noise

INITIALIZATION · NNC

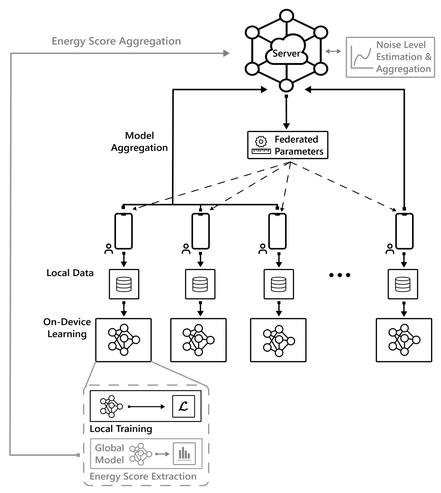


LOCAL TRAINING · AKD



AGGREGATION · NA-FedAvg

Estimate noise once after warm-up



Technical core

$$\mathcal{L}_{\text{AKD}} = \mathcal{L}_{\text{CE}} + \beta u_{\varepsilon}(n_l^m)\mathcal{L}_{\text{KD}}$$

For client m, estimated noise n_l^m gates teacher guidance through u_ε. β controls its strength alongside label-based CE.

$$\gamma_m = (1 - n_l^m)\frac{N_m}{N}$$

NA-FedAvg multiplies client m's data share N_m/N by its estimated clean-label fraction $1\text{-}n_l^m$.

Paper **Code**

03 Evaluation

+17.11 pp
NNC vs. FedAvg
CIFAR-10; 70% random noise

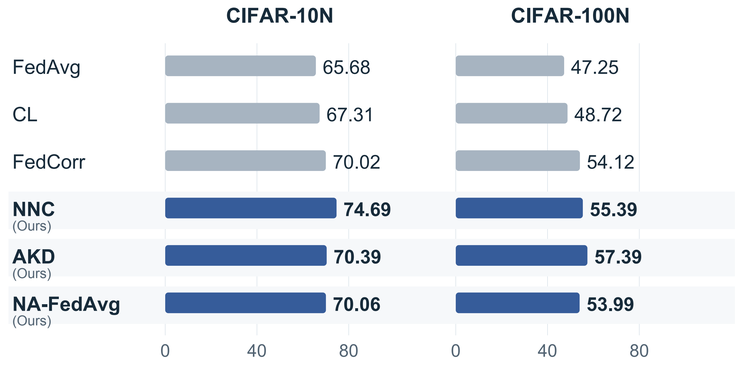
+10.14 pp
AKD vs. FedAvg
CIFAR-100N; human label noise

Accuracy depends on the noise pattern

Method	Random noise (%)	Class flipping (%)
FedAvg	57.65	63.54
FedCorr	67.27	65.23
NNC (Ours)	**74.76**	**54.59**
AKD · embeddings (Ours)	**68.43**	**68.39**
NA-FedAvg (Ours)	**65.34**	**67.92**

CIFAR-10/ResNet-20; 30 clients, 200 rounds, three-run mean accuracy. 70% label noise on 80% of clients; random versus class-flipping noise.

Accuracy with human annotation noise



Three-run mean test accuracy (%); dispersion unreported. 30 clients, 80% participation; 200/500 rounds for CIFAR-10N/100N. *AKD* uses embeddings.

Takeaway

Match the intervention to the noise: *NNC* gains 17.11 pp under random noise but falls below FedAvg under class flipping. With human annotations, **NNC leads on CIFAR-10N and AKD on CIFAR-100N** among these methods.

Privacy-preserving Speech Emotion Recognition through Semi-Supervised Federated Learning

Vasileios Tsouvalas · Tanir Ozcelebi · Nirvana Meratnia

Eindhoven University of Technology

EmotionAware · IEEE PerCom Workshops 2022

01 Motivation

The problem

Emotion annotation is costly, leaving much device speech unlabeled. Speech emotion recognition (SER) also involves sensitive recordings, motivating collaborative training that keeps raw audio on devices.

Contributions

- **Attention-based features** capture spectral, temporal and channel cues from speech.
- **Device-specific confidence** filters pseudo-labels according to training progress and client participation.
- **Fixed-budget comparisons** test four speaker/client splits with only 10% labels.

Local data, no server corpus

Each client has a small labeled subset and more unlabeled speech. The approach requires no server-side training data.

Speaker variation

Random and per-speaker partitions test different client distributions. Seen- and unseen-speaker evaluation separates familiar speakers from speakers absent during training.

Scope

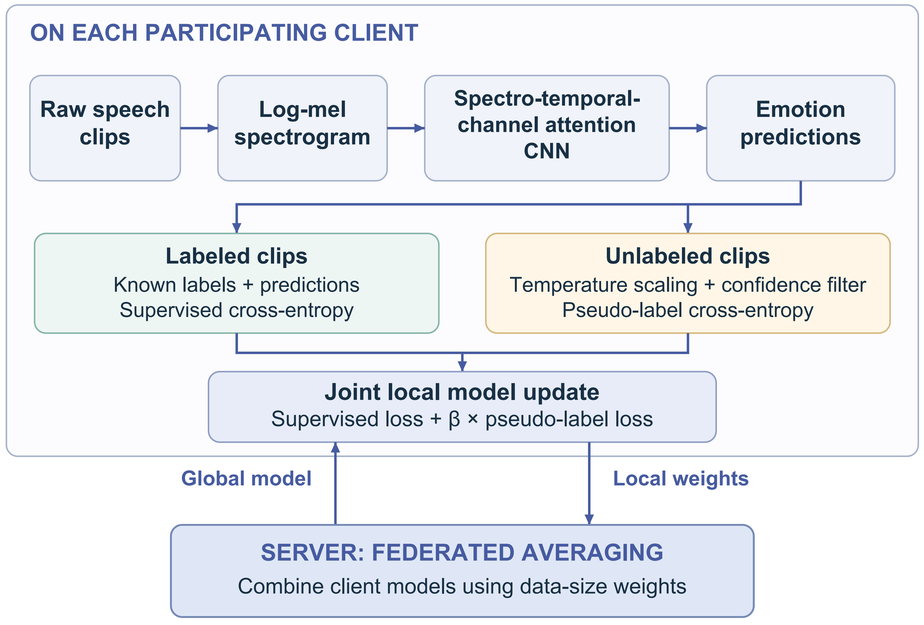
Evidence comes from one four-emotion corpus without reported dispersion. Keeping raw speech local does not provide differential privacy or protection against model-update attacks.

Notation

convolutional neural network (CNN) · federated learning (FL) · cross-entropy (CE)

02 FedSER

Use unlabeled speech without centralizing raw recordings



Technical core

$$\mathcal{L}_k = \mathcal{L}_s(\mathcal{D}_L^k) + \beta \mathcal{L}_u(\mathcal{D}_U^k)$$

Client k combines supervised loss on labeled speech D_L^k with loss on confident pseudo-labels from D_U^k; β sets their relative influence.

Paper

Code

03 Evaluation

10%
labeled speech
Remaining clips used unlabeled

56.83%
unweighted accuracy
Unseen speakers; per-speaker clients

Unlabeled speech helps at a fixed label budget

Speaker/client split	Supervised 10%	Self-training 10% (Ours)	Supervised 100%
Unseen / random	46.72	**53.94**	60.52
Unseen / per speaker	46.55	**56.83**	59.55
Overlap / random	51.58	**59.04**	64.78
Overlap / per speaker	49.97	**59.72**	62.79

IEMOCAP improvised speech, four emotions; five-run mean unweighted accuracy (%). Random/per-speaker partitions use 10/8 clients.

Fully labeled reference models

Same architecture	Unseen speakers (%)	Overlapping speakers (%)
Centralized	62.58	69.04
Federated	60.52	64.78

100% labels, random partitions; mean unweighted accuracy. FL: 10 clients, 100 rounds, 80% participation, one local step.

Takeaway

Self-training improves every 10%-label comparison, while remaining below the fully labeled reference. It keeps **raw recordings local**; the experiments establish no formal privacy guarantee.

Federated Self-training for Semi-supervised Audio Recognition

Vasileios Tsouvalas · Aaqib Saeed · Tanir Ozcelebi

Eindhoven University of Technology, The Netherlands

ACM TECS · Vol. 21, No. 6, Article 74 · 2022

EINDHOVEN UNIVERSITY OF TECHNOLOGY

01 Motivation

The problem

Scarce annotations leave most on-device audio unused by supervised federated learning (FL). The challenge is to learn from this unlabeled audio without collecting users' recordings centrally.

Contributions

- **Joint local learning** uses labeled and unlabeled audio within one model.
- **Confidence filtering** adjusts pseudo-label selection as training progresses.
- **Label-budget evaluation** tests client distributions and optional public-audio pretraining.

Client data assumption

Each client holds few labeled clips and many unlabeled clips, assumed to share its local distribution. Training seeks one shared audio classifier.

Optional initialization

Public unlabeled audio can initialize the model through self-supervised pretraining. Experiments compare this optional initialization with random model weights.

Scope

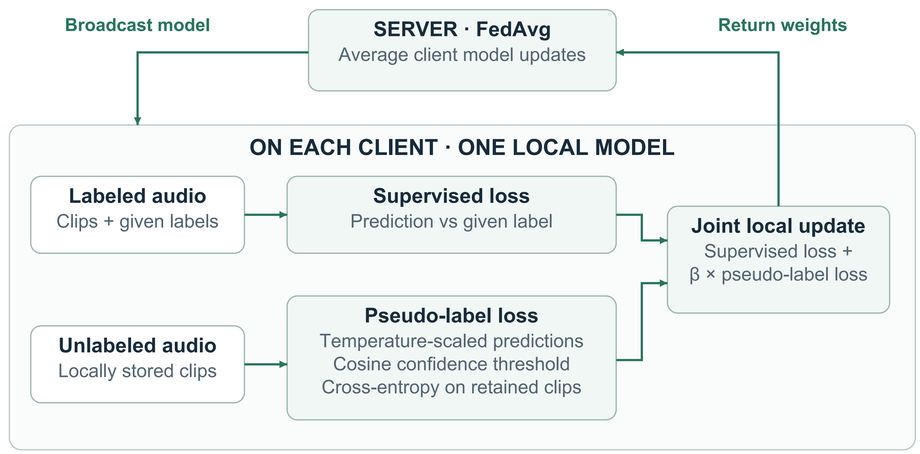
Training uses simulated clients. Class-skew tests keep unlabeled clips randomly distributed, so they do not establish robustness to matching labeled- and unlabeled-data skew.

Notation

cross-entropy (CE)

02 FedSTAR

One local model learns from true labels and pseudo-labels



Compute both losses before the same backward update.

Repeat across selected clients and federated rounds. Raw downstream audio stays local.

Technical core

$$\mathcal{L}_k = \mathcal{L}_{\mathrm{CE}}(y, p_{\theta^k}(x_l)) + \beta \mathcal{L}_{\mathrm{CE}}(\hat{y}, p_{\theta^k}(x_u))$$

One client model uses true labels y and confident pseudo-labels ŷ. β weights the unlabeled-audio loss; both terms share one backward pass.

Paper

Code

03 Evaluation

3%
labeled data
15-client comparison

86.82%
Speech Commands accuracy
FedSTAR at 3% labels

Learning with 3% labels

Dataset	Supervised FL (%)	FedSTAR (%) (Ours)
Ambient Context	33.03 ± 0.002	**49.54 ± 0.005**
Speech Commands	62.98 ± 0.003	**86.82 ± 0.006**
VoxForge	54.26 ± 0.002	**55.82 ± 0.011**

3% labels, 15 clients, 100 rounds, 80% participation. Three-run mean accuracy; printed ± terms are described as variance.

Pretraining across label budgets

Labels	Random init. (%)	Pretrained (%)
3%	85.40 ± 0.008	85.17 ± 0.001
5%	88.52 ± 0.005	88.28 ± 0.002
20%	92.24 ± 0.012	93.67 ± 0.005
50%	92.26 ± 0.005	93.83 ± 0.003

Speech Commands; 50% of remaining unlabeled clips, 15 clients, 100 rounds. Three-run means; ± terms denote reported variance.

Takeaway

At 3% labels, **self-training improves all three tasks**, but gains vary widely. Public-data pretraining can help early convergence, while its **final-accuracy benefit depends on the data budget**.

On the Effectiveness of Invisible Backdoor Attacks in Federated Learning

Mattjis Marinus · **Vasileios Tsouvalas** · Nirvana Meratnia

Eindhoven University of Technology, The Netherlands

UbiSec 2024 · Proceedings published 2025

01 Motivation

The problem

Invisible backdoor triggers must survive aggregation with honest updates. This study asks when existing attacks remain effective in federated learning (FL) and whether low-activation placement improves success without making detection easier.

Contributions

- **Federated comparison:** Evaluate four invisible attack families against visible BadNet triggers.
- **Multiple metrics:** Separate attack success, clean accuracy, visual similarity and model-level detection.
- **Placement experiment:** Test proposed low-activation placement with least significant bit (LSB) triggers and matched controls.

Attacker setting

A minority of clients poisons local training samples toward a chosen target class while honest clients train normally. The server aggregates both. The study varies attacker prevalence and compares selected defenses.

Success is not stealth

Attack success rate (ASR) measures triggered misclassification; clean accuracy measures retained task performance. Learned Perceptual Image Patch Similarity (LPIPS) compares image appearance, whereas Neural Cleanse tests model anomalies. A visually subtle trigger can still be detected.

Scope

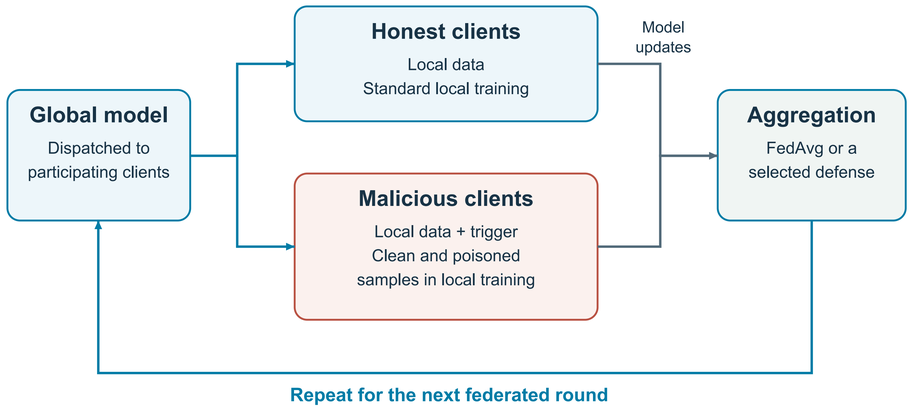
Experiments use ResNet-18 and small image datasets; repeated-run counts and the meaning of ± are unspecified. Placement evidence covers LSB.

Notation

Dirichlet distribution with concentration a (Dir(a)) · percentage points (pp)

02 FiBA

Test whether invisible triggers survive aggregation



Technical core

$$\mathcal{L}_{\text{attack}} = \mathcal{L}_{\text{clean}} + \mathcal{L}_{\text{poison}\to t}$$

The two cross-entropy terms fit clean labels and poisoned samples labelled as target class t. Honest updates compete with this objective.

$$\text{ASR}(\%) = 100\,\frac{N_{\text{triggered}\to t}}{N_{\text{triggered}}}$$

$N_{\text{triggered}}$: poisoned test samples excluding the target class; the numerator counts those predicted as target t. ASR measures attack success.

Paper

Code

03 Evaluation

88.59%
Encoder attack success
20% attackers · CIFAR-10

+19.80 pp
LSB700 ASR difference
Low minus unmasked placement

Attacker prevalence changes attack success

Attack	Clean acc., 10%	ASR, 10%	ASR, 20%
BadNet	84.00 ± 0.31	94.96 ± 0.81	97.52 ± 0.40
LSB	83.37 ± 0.33	41.59 ± 6.19	65.43 ± 3.93
WaNet	83.33 ± 0.31	44.41 ± 3.50	82.05 ± 3.59
Encoder	83.44 ± 0.27	29.44 ± 5.90	88.59 ± 4.01
LIRA	83.71 ± 0.41	10.52 ± 1.22	41.57 ± 4.66

CIFAR-10 · ResNet-18 · FedAvg · 10 clients · 100 rounds · Dir(0.5). Poison rate 0.5; target class 5. Columns give attacker fraction; reported ± is undefined.

Low-activation placement has a cost

LSB700 placement	ASR (%)	LPIPS	Anomaly index
Unmasked	41.59 ± 6.19	7.83e-5	1.61
Low (Ours)	**61.39 ± 4.47**	**2.25e-4**	**3.04**
High	36.89 ± 4.26	3.75e-4	2.01

LSB700 · CIFAR-10 · 10% attackers. Lower LPIPS means closer images; Neural Cleanse flags anomaly above 2. “Low” is the proposed placement.

Takeaway

Encoder reaches **88.59% ASR with 20% attackers**, showing the importance of attacker prevalence. Proposed **low-activation LSB700 placement** gains 19.80 points over unmasked placement, but raises Neural Cleanse’s anomaly index from 1.61 to 3.04.

Many-Task Federated Fine-Tuning via Unified Task Vectors

Vasileios Tsouvalas · Tanir Ozcelebi · Nirvana Meratnia

Eindhoven University of Technology

FedGenAI @ IJCAI 2025

01 Motivation

The problem

Many-task federated learning (FL) must share knowledge without forcing dissimilar tasks into one update. Separate per-task models increase storage and communication, while clients may hold several tasks simultaneously.

Contributions

- **Compact representation:** combine task vectors with lightweight masks and scalar rescalers.
- **Related-task transfer:** aggregate within tasks and across tasks with aligned update signs.
- **Stateless server:** avoid retaining client-specific model weights between rounds.

Beyond one task per client

Clients can hold multiple overlapping tasks, each with its own data. This task heterogeneity creates interference within clients as well as across clients.

A shared reference point

Task vectors measure changes from the same pretrained weights. Conflicting update signs indicate potential interference; agreement provides a data-free signal of task compatibility.

Scope

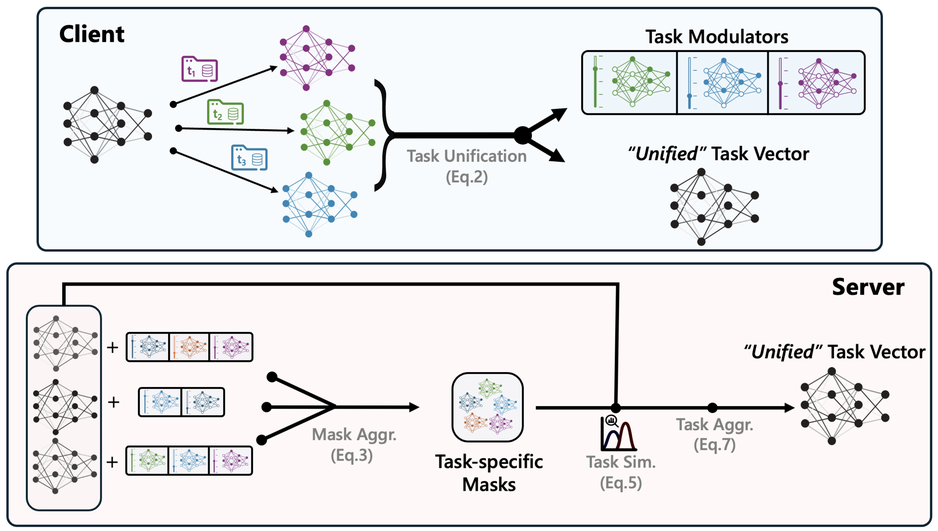
Eight-task comparisons use unequal round budgets; repeated-run uncertainty is unreported. Evidence is limited to vision classification with vision transformer (ViT)-B/32 and low-rank adaptation (LoRA).

Notation

megabits (Mbit)

02 MaTU

One vector per client, distinct updates per task



Technical core

$$\tau = \sigma \odot \mu$$

τ: unified task vector; σ: sign of the summed task vectors; μ: largest magnitude agreeing with that sign, entry by entry.

$$\dot{\tau}_t = \lambda_t m_t \odot \tau, \qquad \theta_t = \theta_p + \dot{\tau}_t$$

Mask m_t keeps task-aligned entries; scalar λ_t rescales them. Add the recovered task vector to pretrained weights θ_p.

Paper

Code

03 Evaluation

79.47%
average task accuracy
8 tasks; multiple tasks per client.

8.04 Mbit/task
communication per round
MaT-FL: 20.43 Mbit/task/round.

Multiple tasks per client: accuracy and communication

Method	Avg. accuracy (%)	Mbit/task/round
Individual centralized	90.21	—
FedAvg	51.63	20.43
FedPer	48.37	17.16
MaT-FL	67.60	20.43
MaTU (Ours)	**79.47**	**8.04**

8 tasks; ViT-B/32, LoRA rank 16. 30 clients, 1 local epoch, task concentration ζt=0.5, 20% participation. 100 rounds; MaT-FL: 300. No uncertainty reported.

Scaling to the 30-task benchmark

Method	Normalized accuracy (%)
MaT-FL	52.62
MaTU (Ours)	**77.40**

30 tasks; accuracy normalized to individual task fine-tuning. ViT-B/32, 30 clients, 300 rounds, 1 local epoch, task concentration ζt=0.2, full participation.

Takeaway

MaTU reaches **79.47% at 8.04 Mbit/task/round** across eight tasks. Its 30-task normalized accuracy is **77.40% versus MaT-FL's 52.62%**, below individual task fine-tuning.

Fine-tuning Multimodal Transformers on Edge: A Parallel Split Learning Approach

Timo Fudala · **Vasileios Tsouvalas** · Nirvana Meratnia

Eindhoven University of Technology

FedGenAI @ IJCAI 2025

01 Motivation

The problem

Multimodal transformers exceed typical edge training budgets. Parameter-efficient fine-tuning (PEFT) in federated learning (FL) reduces trainable parameters but still executes the backbone on each client, so fewer updated parameters do not necessarily mean less client computation.

Contributions

- **Multimodal splitting:** extend parallel split learning (PSL) with one shared modality-agnostic encoder.
- **Shared optimization:** aggregate local losses for one server backward pass, keeping raw labels local.
- **Cross-task evidence:** evaluate performance and client cost across seven paired-modality datasets.

Paired inputs, distinct tasks

The study covers vision-text, vision-audio and audio-text pairs. Classification and retrieval impose different fusion and modality-alignment demands.

Where compute is measured

Client compute is measured in billions of floating-point operations (GFLOPs) per input. Offloading shifts computation to the server, so this metric alone does not describe total-system cost.

Scope

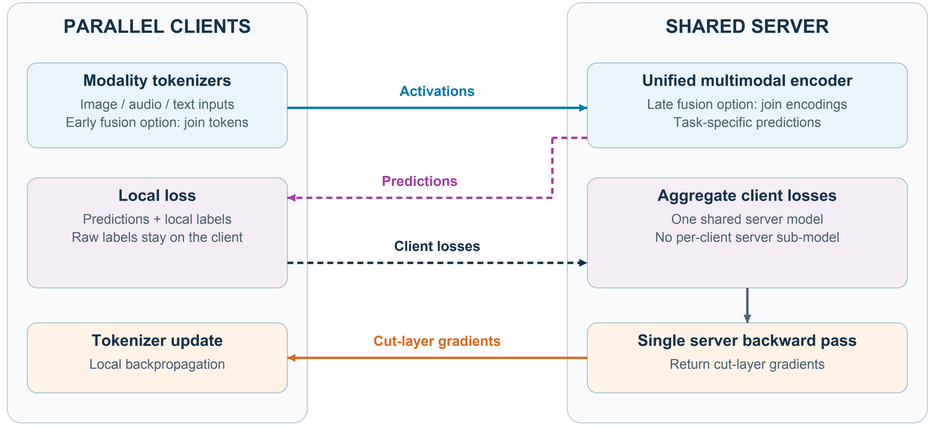
Evaluation averages client tokenizers into one model. Missing-modality robustness is untested; classification gains vary by task, and retrieval remains below centralized fine-tuning.

Notation

multimodal parallel split learning (*MPSL*) · Dirichlet distribution (Dir) · vision transformer (ViT) · recall at rank 1 (R@1)

02 MPSL

One shared encoder, lightweight clients



Technical core

$$\mathcal{L}_S = \sum_{n \in \mathcal{N}} \frac{|\mathcal{B}_n|}{|\mathcal{B}|} \mathcal{L}_{C_n}$$

L_{Cn}: local client loss; B_n: client batch; B: combined batch. Minimize this batch-weighted loss in one server backward pass.

Paper

Code

03 Evaluation

0.1 GFLOPs
average client compute
Per input, five classification tasks; FedAvg: 25.8.

58.2 ± 0.9%
MELD accuracy
100 clients; FedAvg: 48.9 ± 0.0%.

Client efficiency with task-dependent accuracy

Method	COCO-QA	UCF101	MELD	Client GFLOPs
Centralized	61.7 ± 0.1	91.3 ± 0.2	61.4 ± 0.4	25.8
FedAvg	53.5 ± 0.2	89.8 ± 0.9	48.9 ± 0.0	25.8
FedCLIP	13.1 ± 2.6	56.4 ± 1.2	50.7 ± 0.7	25.8
MPSL (Ours)	**46.2 ± 0.5**	**89.3 ± 1.0**	**58.2 ± 0.9**	**0.1**

Accuracy (%), reported ±, 3 seeds. Distributed methods: 100 clients, Dir(0.1); Meta-Transformer ViT-B/16, last 6 blocks trained. Client GFLOPs/input average five classification tasks.

Image-to-text retrieval remains challenging

Method	MS-COCO R@1	Flickr30K R@1	Client GFLOPs
Centralized	56.8 ± 2.4	66.8 ± 5.3	8.65
FedAvg	1.5 ± 0.1	11.2 ± 1.7	8.65
FedCLIP	0.1 ± 0.0	0.1 ± 0.0	8.65
MPSL (Ours)	**21.1 ± 0.8**	**26.0 ± 0.5**	**0.1**

Image-to-text R@1 (%), reported ±, 3 seeds. Distributed methods: 100 clients; full ViT-B/16 encoder fine-tuning. GFLOPs are per client input.

Takeaway

MPSL lowers average client compute to **0.1 GFLOPs per input**, with a MELD gain and COCO-QA deficit versus FedAvg. Retrieval improves over the tested federated baselines but remains below centralized training.

Plug-and-Play Multilingual Few-shot Spoken Words Recognition

Aaqib Saeed · **Vasileios Tsouvalas**

Eindhoven University of Technology, The Netherlands

arXiv:2305.03058 · 2023

01 Motivation

The problem

Fixed-vocabulary keyword spotting (KWS) requires new training when users add words or languages. Low-resource languages and personal vocabularies need recognition from only a few recordings.

Contributions

- **Multilingual embeddings** learn from over 12 million spoken-word clips across 20 languages.
- **Unseen-class evaluation** tests new words, five new languages and language-identification transfer.
- **Edge measurements** compare small and base encoders across three devices.

Few-shot episodes

An episode contains N candidate words and K labeled support clips per word. Queries are separate clips from those same classes; training and test vocabularies are disjoint.

Closed-set recognition

Each query belongs to one candidate word in the support set. Experiments use isolated clips; continuous listening and unknown-word rejection are outside this evaluation.

Scope

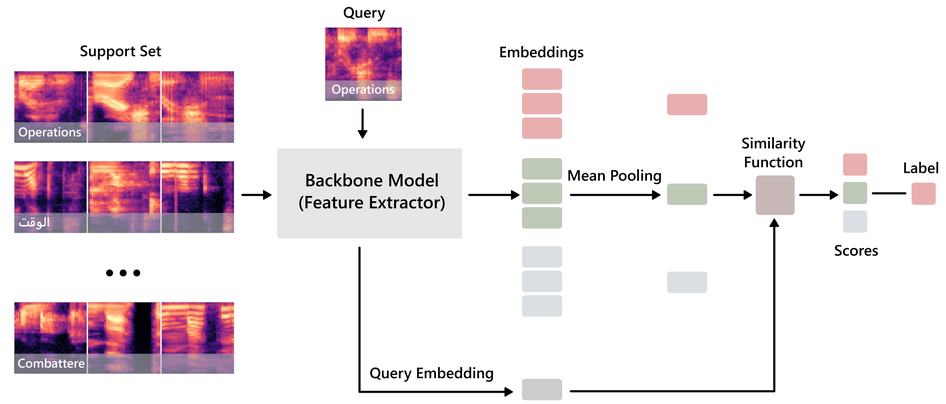
Encoder latency excludes preprocessing and classification. Recognition accuracy after quantization is not reported.

Notation

Multilingual Spoken Words Corpus (MSWC) · few-shot learning (FSL)

02 PLiX

New examples add vocabulary without retraining the encoder



Technical core

$$\mathbf{p}_i = \frac{1}{|S_i|} \sum_{(x,y) \in S_i} f_\theta(x)$$

S_i contains support examples of word i; f_θ encodes each clip. Their mean p_i defines that word without updating θ.

$$\hat{y} = \arg\min_i \| f_\theta(x_q) - \mathbf{p}_i \|_2^2$$

Encode query x_q separately and choose the closest word prototype. Query embeddings are never averaged into p_i.

Paper Code

03 Evaluation

95%
80-word recognition
English-only base; five supports/word

5.12 ms
small encoder latency
Quantized TinyNet-E; Pixel 7; encoder only

More support handles larger vocabularies

English test vocabulary	Supports per word	Accuracy (%) (Ours)
40 words	1	**90**
80 words	1	**86**
60 words	5	**96**
80 words	5	**95**

English-only base model, held-out MSWC words; 1,000 episodes. Rounded mean accuracies.

Quantized encoder latency

Device	Small (ms)	Base (ms)
Pixel 7	5.12	78.07
Raspberry Pi 4	6.92	107.13
Jetson Nano	7.32	101.37

Quantized TinyNet-E / EfficientNet-v2 Medium; 200-run mean encoder latency. Excludes log-mel preprocessing and prototype classification.

Takeaway

New support examples add vocabulary without encoder retraining, with accuracy depending on language, vocabulary size and support count. The **5.12 ms** Pixel 7 result measures only the separate small encoder, excluding preprocessing and classification.